Krishnaa **Vadivel** | Machine Learning Engineer | Scientific ML | Equivariant Models

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Summary

Machine-learning-focused PhD researcher working at the intersection of scientific computing, molecular/materials modeling, and scalable infrastructure. Experience spans PyTorch and equivariant graph models, agentic retrieval workflows, GPU acceleration, and distributed compute on national HPC systems.

Experience

Yale University, Electrical Engineering
PhD Research Assistant *2021–present*
- Develop scientific ML workflows around first-principles materials research, including equivariant molecular models, physics-informed formulations, and ML-assisted acceleration of computational pipelines.
- Build distributed and accelerator-enabled workflows in Python and C++ using CUDA, ROCm, MPI, OpenMP, and HPC environments such as Frontier and Perlmutter.
- Explore theory-to-implementation links across graph neural networks, diffusion models, transformer-style abstractions, and data generation for scientific use cases.

Probe Technologies Inc. / Weatherford Wireline Services
Software and Embedded Systems Engineer *2019–2021*
- Built GPU-accelerated processing and custom OpenGL visualization tools for large waveform datasets, providing practical experience with high-throughput data pipelines and performance-oriented engineering.
- Supported production data and engineering workflows with ASP-style services and Microsoft SQL-backed tooling.

Selected Projects

Equivariant Molecular ML: PyTorch and graph-based workflows for molecular-property prediction and symmetry-aware learning on QM9-style data.

Agentic Scientific RAG: Retrieval-driven tooling that reads documentation and research papers to generate structured inputs and workflow scaffolds for computational science.

Diffusion and Transformer Theory Notes: Practical derivations and organized notes for DDPM-style models, conditioning, and index-notation ML formulations.

Distributed Scientific Compute: ML-adjacent acceleration work combining CUDA/ROCm, HPC scheduling, and scalable simulation pipelines.

Education

Yale University
PhD, Electrical Engineering *2021–present*

Yale University
MPhil, Electrical Engineering *2023*

Yale University
MS, Electrical Engineering *2023*

Cornell University
MEng, Electrical and Computer Engineering *2017–2018*

Cornell University
BS, Electrical and Computer Engineering *2013–2017*

Technical Skills

ML: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks
Programming: Python, C++, CUDA, JavaScript, Bash
Data / Compute: NumPy, pandas, SciPy, distributed workflows, CUDA, ROCm
Systems: Linux, Docker, Docker Compose, Kubernetes, gcloud
Scientific: PETSc, SLEPc, scientific simulation pipelines, physics-informed modeling